# HENRIQUE SCHECHTER VERA

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## EDUCATION

**Princeton University B.S.E. in Computer Science** *Sep. 2019–May 2023*
**Honors:** Summa Cum Laude, Phi Beta Kappa (early inductee), Tau Beta Pi (early inductee)
Minors in Applied & Computational Mathematics, Statistics & Machine Learning, Optimization & Quantitative Decision Science
**Overall GPA: 3.99, Departmental GPA: 4.0**

- **Graduate coursework:** Advanced Algorithm Design, Data Structures and Graph Algorithms
  **Undergraduate:** Natural Language Processing (NLP), Machine Learning Theory, Machine Translation, Optimization
- **Natural languages:** (native/fluent proficiency) English, Spanish, Portuguese, Guarani
- **Programming languages:** Java, Python, C, C++, Go, SQL, HTML, CSS, JavaScript, Assembly, R, MATLAB

## EXPERIENCE

**Google DeepMind**, *Senior Research Engineer* *November 2025–Present*

- Conducting fundamental research in multimodal representation learning, computer vision, and language understanding
- Leading cross-functional teams from research through productionization and GTM

**Google**, *Senior Machine Learning Engineer* *July 2023–November 2025*

- Drove research in natural language understanding and landed state-of-the-art innovations across Alphabet products
- Worked on the research, development, and deployment for Google Search's query understanding models (XXXB searches/day)

**Google**, *Software Engineering Intern* *May 2022–August 2022*

- Completely re-implemented team's flagship neural semantic parsing model
- Designed architecture alongside researchers, implemented the model, and built training and evaluation infrastructure
- Led initiative and communication with research teams to open-source our model's classifier and new debugging tools

**Google**, *Software Engineering Intern* *May 2021–August 2021*

- Developed end-to-end features for electric vehicles in Google Maps, using NLP to create voice queries for Google Assistant
- Served as direct liaison between two teams involved, driving communications with UX, PM, and SWE correspondents
- Went beyond assigned tasks: Successfully led work with a fellow intern to jumpstart a related project
- Fixed a year-old bug directly affecting user experience. Answered several intern questions on Q&A site and groups

**Landed (Y Combinator W19)**, *Software Engineering and Product Management Intern* *June–Aug. 2020*

- Created and shipped standalone, customer-facing pricing tool using JavaFX and CSS. Determines cost based on amount, location, urgency, and job of the hires. Continues to be used in meetings with prospective clients
- Led team through product development life cycle for various projects, from ideation to deployment to analysis
- Introduced successful features to platform with 10k+ users, such as a preference-based jobs page and filming overlay tips
- Created data analytics dashboards and tracked key metrics to identify growth opportunities, evaluate product performance, and deliver concrete insights. Used SQL (PostgreSQL), Excel, and Google Data Studio

**Princeton Computer Science**, *Research TA, Head TA, Instructor, Grader, Tutor, Interactor* *Feb. 2020–2023*

- Analysis of Succinct Data Structures (Research TA), Machine Translation (Head TA), Machine Learning (TA + grader), Reasoning About Computation (Tutor), Algorithms and Data Structures (TA + grader), Programming Systems (TA), Introduction to Computer Science (Instructor + TA + Grader)
- Research TA: help students with CS research projects, give feedback for proposals and papers (with Prof. Huacheng Yu)
- Head TA: Created and led part of curriculum for intro NLP course, held office hours, monitored discussion forum
- Instructor: Led an introductory CS class twice a week, teaching 15 Princeton undergraduates
  Evaluated applications for future Undergraduate Course Assistants
- Interactor: Worked with a professor as academic advisors for 1st-year CS majors

**Po Paraguay**, *Product and UX Intern* *June 2017*

- Conducted user interviews and tested prosthetic performance to refine product designs
- Created public product pages for our 3D-printed prosthetics, with multilingual assembly guides and videos of prosthetic functionality, increasing reach and accessibility. Recruited and managed 60 volunteers for 2 years afterward

## RESEARCH

**EmbeddingGemma 2: Unified and Efficient Multimodal Representations** 2026
*Google DeepMind (Paper)*
Best open on-device multimodal embedding model, unifying modalities and combinations thereof into one representation space

**Gemini Embedding 2: A Native Multimodal Embedding Model from Gemini** 2026
*Google DeepMind (Paper)*
State-of-the-art flagship API model encoding interleaved text, image, audio, and video, surpassing even specialist models

**EmbeddingGemma: Powerful and Lightweight Text Representations** 2025
*Google DeepMind (Paper)*
Compact 300M-parameter open text embedding model matching the best models more than twice its size. 25M+ downloads

**Gemini Embedding: Generalizable Embeddings from Gemini** 2024–2025
*Google DeepMind (Paper)*
Multilingual API model using LLMs for synthetic data and hard-negative mining to reach state-of-the-art text and code quality

**Structured Pruning for Pre-Trained Models** 2022–2023
*Adviser(s): Prof. Danqi Chen, Mengzhou Xia (Princeton Language and Intelligence) (github.com/princeton-nlp/CoFiPruning)*
Structured pruning method jointly masking coarse and fine-grained modules, alongside a dynamic layerwise distillation strategy

**Selling to a Sophisticated No-Regret Buyer** 2022–2023
*Adviser(s): Prof. Matt Weinberg, Second Reader: Prof. Mark Braverman (Paper)*
Provably near-maximal revenue auctions for buyers using no-regret online learning with recency-bias and $k$-switching

**Extending Randomized Single Elimination Bracket to Multiple Prize Vectors** 2022–2023
*Adviser(s): Prof. Matt Weinberg, Prof. Huacheng Yu (Paper)*
Two novel multi-prize tournament ranking rules with proven guarantees for fairness and manipulability

**Exploring the Role of Contextualization in Dynamic Contextualized Word Embeddings (DCWEs)** 2022
*Adviser(s): Prof. Karthik Narasimhan (Paper)*
Study showing temporal-social embeddings do not aid most non-contextual or contextual language models in sentiment analysis

**Cannabis Analysis: Understanding Strain Names through NLP** 2021–2022
*Adviser(s): Prof. Christiane Fellbaum, Prof. Jérémie Lumbroso (Paper)*
Largest public cannabis strain dataset and NLP study showing lineage, not name sentiment, drives popularity

**Lyric-Based Automatic Playlist Continuation Using Feature Engineering** 2021
*Adviser(s): Prof. Srinivas Bangalore (Paper)*
Neural playlist continuation system using engineered lyric features on 1M Spotify playlists, beating the baseline by 4x

## ACTIVITIES

**Princeton Data Science**, *Officer* *2021–2022*
- Coordinated educational student events and speaker talks with experts, and established formal partnerships with companies

**Princeton Puzzle Hunt**, *Web Developer* *2020–2022*
- Learned HTML, CSS, and JavaScript to help reconstruct and manage the entire website for this fully digital event

**Jane Street ETC; Facebook; HackPrinceton; ReHack.co**, *Competitor; Fellow; Coder; Team Captain* *2019–2021*
- Created an algorithmic trading bot. Learned finance and socket programming. Placed 6th out of 22 teams
- Learned advanced CS over 10 weeks with FB Software Engineers
- Built privacy app that utilizes ML-based facial recognition to protect one's computer from unwanted users

**Tigers For Nassau, Global Research and Consulting Group**, *Technology Strategist and Consultant* *2020–2021*
- Created digital solutions (delivery platforms, ordering systems) to adapt businesses to the COVID-19 crisis

**TechTogether New York**, *Organizer* *2020*
- Coordinated and carried out corporate deals and partnerships. Designed our sponsorship package

**Student Government and Student Council**, *Class Representative, Class Officer (Treasurer)* *2016–2019*
- Supervised marketing, partnerships, finances, staffing, and logistics for events with 40,000+ attendees and 1,500+ athletes

## AWARDS

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| **Phi Beta Kappa** | Early Inductee, Sep. 2022 |
| **Tau Beta Pi** | Early Inductee, Sep. 2021 |
| **Paraguay National Math Olympiad** | (2x) Silver Medal and Departmental Champion (2017–2018) |
| **International Math Kangaroo** | 1st, 3rd, 3rd, 5th, 5th, and 7th place nationally (2010–2019) |